

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
 Subject Code – EEPC203
 Maximum Marks: 50

Programme: B.Tech

Subject: Generation Transmission and Distribution

Total No. of Questions 05 (one question has internal choice)

Semester 3rd

Total No. of Pages used: 02

Number of Questions to be attempted: 05

Time allowed: 03 Hours

Note:

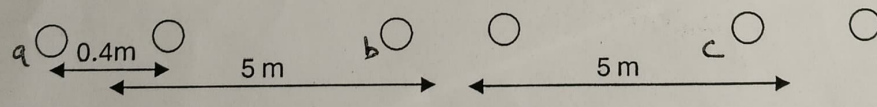
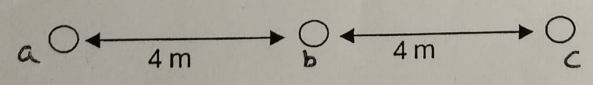
- Unless stated otherwise, the Symbols have their usual meanings in context with the Subject.
- Assume suitably and state, additional data required, if any.
- Please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of right course no. and right subject title.
- All the parts of questions must be answered at the same place.
- Question -1 has internal choice.

Q. No.	Question	Marks	CO
1. (a)	Explain the operation of a steam power plant using a well labeled block diagram.	5	CO1
1. (b)	The load curve of an electrical system is linear with the following values at different times of the day:	5	CO1

Time	12 am	5 am	9 am	6 pm	8 pm	10 pm	12 am
Load (MW)	50	50	100	100	150	80	50

Plot the chronological load curve. Find the energy required by the system in one day and the system daily load factor.

OR

1. (a) What is the significance of a) Maximum Demand, b) Load Factor, c) Base load plants, d) Peak load plants, and e) diversity factor
1. (b) Explain the operation of a hydro power plant using a well labeled block diagram.
2. (a) Find the inductive reactance in ohms/kilometer at 50 Hz of a 3-phase bundle conductor line with two conductors per phase as shown in figure below. radius of conductor is 2 cm.
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2. (b) Determine the capacitance and charging current per km when a line shown in figure operating at 132 kV and having conductor diameter 1.0 cm.
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3. (a) A 3-Phase, 50 Hz line is operating at the corners of an equilateral triangle. 220 kV which has conductor of 1.5 cm diameter arranged in 3.0 meters apart. Air temperature and pressure are 200°C and 75 cm Hg Height respectively. The Dielectric strength of air

and irregularity factor are 21 kv / cm and 0.85 respectively. Determine the critical disruptive voltage and corona loss.

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| 3. | (b) | Determine the efficiency of an insulator string having four disc. The capacitance of each disc is nine times of the capacitance to ground. The operating voltage of line is 33 kV line to ground. | 5 | CO4 |
| 4. | (a) | Explain the construction of a single core and three-core cable. Discuss the limitations and advantages of using cables for delivering electric power. | 5 | CO4 |
| 4. | (b) | What are the different methods of Grading in Cables. Analyze the performance of any one of these methods of cable grading. | 5 | CO4 |
| 5 | (a) | Short answer Questions | 2 x 5= 10 | |
| | | 1) Briefly explain the Ferranti Effect and Skin Effect | | CO2 |
| | | 2) Compare the AC and DC Distribution Systems | | CO3 |
| | | 3) Compare the Radial, Ring and Mesh Distribution Networks | | CO3 |
| | | 4) Analyze the conductor material required for a Three phase four wire AC distribution system | | CO3 |
| | | 5) Analyze the conductor material required for a Two wire DC distribution system | | CO3 |

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